Testing the tectonic significance of river-profile and drainage-basin morphometry in low-strain intraplate settings: a comparative review from the Ardenne, the western Rhenish shield and the Upper Rhine Graben (NW Europe)

First Author[a,*] and Second Author[b]

[a] *Department of Geology, Name of University, City – PIN, India*
[b] *Department / Institute, Name of University, City – PIN, India*
[*] *Corresponding author. E-mail: corresponding.author@example.com*

Abstract

Rivers in slowly deforming regions carry only faint traces of tectonics, and these traces are easily confused with the effects of rock strength, discharge, inherited relief and Pleistocene climate. This review compares three studies that tested how far morphometric indices can isolate a tectonic signal in NW Europe. Demoulin (1998) analysed the long profiles of 24 rivers in the Ardenne using map-based measures of gradient, concavity and valley-side height. Peters and Van Balen (2007) combined palaeogeographic reconstruction, fault–drainage overlays and four geomorphic indices along the borders of the northern Upper Rhine Graben. Demoulin (2011) built a new index, R_1, from the hypsometric integrals of the basin, its drainage network and its trunk stream for 85 rivers of the western Rhenish shield. Re-plotting their published data shows a common strategy: each study first removes the dominant non-tectonic control (hydrodynamics, lithology or basin size) and then interprets what remains. Mean river gradient in the uplifted NE Ardenne (12.4‰) is two to three times that of the western massif and the forelands. Active and inactive segments of the graben border faults separate clearly in the joint behaviour of mountain-front sinuosity (1.1–2.1) and valley shape (1.5–19.9). The slope S_r of R_1 against the logarithm of basin area falls from 0.51 to 0.24 from the youngest to the oldest uplift area. Profile indices therefore show where tectonics acts over 10^4–10^5 years, whereas S_r gives the relative timing of uplift over 10^5–10^6 years. We discuss the shared limits of these methods (lithology, inheritance, data resolution and the lack of absolute ages) and propose a multi-scale workflow for low-strain settings.

Keywords: Longitudinal river profile; Stream-gradient index; Mountain-front sinuosity; Hypsometric integral; R_1 and S_r indices; Intraplate uplift; Ardenne–Rhenish shield; Upper Rhine Graben

1. Introduction

Rivers respond to uplift and faulting by changing their slope, their long-profile shape and the depth of their valleys, and these changes can be measured from maps and elevation models. Since the early work on stream gradients (Hack, 1973) and on mountain fronts and valley shapes (Bull and McFadden, 1977), a large family of morphometric indices has been used to map active tectonics (Keller and Pinter, 2002; Burbank and Anderson, 2001; Kirby and Whipple, 2012). Most of these tools were, however, developed and calibrated in fast-deforming settings such as California, Taiwan or the Betic Cordillera, where uplift rates reach millimetres per year and the tectonic signal is strong (Merritts and Vincent, 1989; Silva et al., 2003; El Hamdouni et al., 2008).

Intraplate NW Europe poses a harder problem. Quaternary uplift and subsidence rates there are mostly between a few hundredths and a few tenths of a millimetre per year. The landscape has been reshaped by repeated glacial–interglacial cycles, it contains planation surfaces inherited from the Tertiary, and the bedrock varies from soft marls to quartzites. Under these conditions, a change in river gradient or valley shape may reflect lithology, a change in discharge or sediment load, a river capture or a periglacial base-level fall just as easily as fault movement. The central question is therefore not only whether tectonics is recorded in the landscape, but whether its record can be separated from everything else.

Three studies from the Ardenne, the western Rhenish shield and the northern Upper Rhine Graben (URG) approach this question from different directions. Demoulin (1998) tested a set of long-profile parameters on 24 rivers of the Ardenne, using only data from 1:25,000 topographic maps. Peters and Van Balen (2007) examined how faulting has shaped the relief and drainage of the northern URG through a palaeogeographic synthesis, an overlay of faults and drainage, and four geomorphic indices. Demoulin (2011) designed a composite index, R_1, that combines the hypsometric integrals of basin, drainage network and trunk stream, and showed that the slope of its relation with basin size records the relative age of uplift.

Taken together, the three papers cover more than a decade of methodological development and three neighbouring structural settings: a Variscan massif undergoing broad uplift, an active rift with well-mapped faults, and the wider Rhenish shield. No synthesis of these studies is available. This review aims to (i) compare their settings, data and indices; (ii) bring their main results together, using the published values to redraw key relationships; (iii) identify the common logic and the shared limits of their methods; and (iv) propose a workflow for future work in low-strain regions.

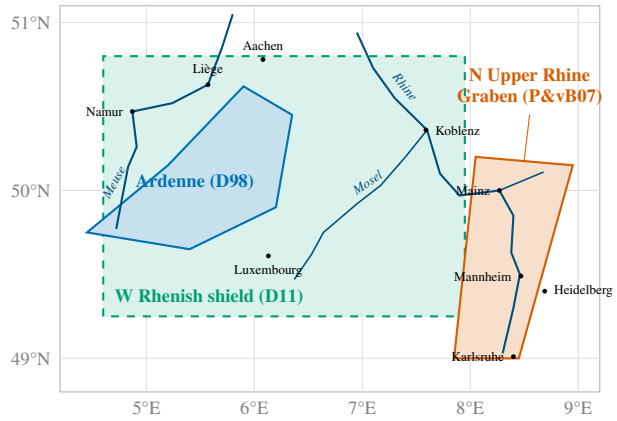


Figure 1. Schematic location of the three study areas (boundaries simplified from the maps in the original papers; rivers generalised). D98: Demoulin (1998); P&vB07: Peters and Van Balen (2007); D11: Demoulin (2011).

2. Scope and approach of the review

The three papers were read in full and compared under five headings: tectonic setting, data and resolution, indices and statistical treatment, main results, and stated limitations. All figures in this review are new drawings based on numbers given in the tables and text of the original papers; no published figure is reproduced. As a check on transcription, the 24 river gradients and bedload proxies listed by Demoulin (1998) were refitted: they return the published regression (gradient $= 7.467\,x - 0.017$) and correlation coefficient ($r = 0.83$) exactly. Values that we calculated, such as ratios between regional means, are labelled as derived.

3. Tectonic and geomorphic setting

3.1. Ardenne massif

The Ardenne is a Palaeozoic massif belonging to the external Variscan belt and forms the western end of the Rhenish shield (Fig. 1). Its basement is cut by ENE–WSW folds and thrusts and by younger NW–SE to NNW–SSE faults, which produces strong contrasts in rock resistance over short distances (Demoulin, 1998). Oligocene marine sands now lying at about 700 m on the Hautes Fagnes plateau show at least 400–500 m of uplift of the NE Ardenne since the Oligocene, against less than 200 m in the west. The Quaternary uplift rate is estimated at about 0.1 mm a^{-1} in the NE and no more than 0.025 mm a^{-1} in the west and south. Some NNW–SSE faults linked to the Lower Rhine rift moved by up to 7 m in the last 100 ka, and the area has a moderate seismicity with one historical event of about M_L 6 (Demoulin, 1998). Annual rainfall ranges from about 800 mm in the forelands to more than 1400 mm on the highest summits.

3.2. Western Rhenish shield

The study area of Demoulin (2011) covers about 240 × 170 km^2 west of the Rhine, from the Ardennes and Eifel in the north to the Hunsrück, the Saar–Nahe basin and the Palatinate in the south. The bedrock is largely slaty and of fairly uniform erodibility. Deformed Tertiary planation surfaces indicate about 450 m of rock uplift since the Oligocene. Fluvial terraces show that uplift accelerated in two steps during the Quaternary, peaking at about 0.5 mm a^{-1} in the NE Ardennes and Eifel between about 0.73 and 0.4 Ma (Van Balen et al., 2000; Demoulin and Hallot, 2009). An uplift wave appears to have moved northwards across the massif since the early Pleistocene, towards the northern Ardennes and Eifel, where uplift is strongest today (Demoulin and Hallot, 2009). This allowed three sub-areas of different uplift age to be defined (Section 5.3).

3.3. Northern Upper Rhine Graben

The URG is the central segment of the European Cenozoic Rift System. Its main rifting phase took place in the Oligocene and Miocene, and subsidence has continued at a lower rate since then (Peters and Van Balen, 2007). Quaternary sediments reach about 380 m in the Heidelberger Loch next to the Eastern Border Fault, where subsidence peaked at about 0.2 mm a^{-1}; elsewhere rates are about an order of magnitude lower. An average deformation rate of 0.03 mm a^{-1} means that one metre of relief takes about 30 ka to build, a span long enough for erosion to remove it. The western graben shoulder has been uplifted relative to the subsiding east, and the valley floor has been heavily modified by human activity. Unlike the two massifs, the URG therefore combines well-mapped faults with very slow, localised deformation.

4. Morphometric approaches compared

Table 1 summarises the three studies and Table 2 defines their indices. Fig. 3 places the indices on an indicative time axis.

4.1. Long-profile geometry and a bedload proxy (Ardenne)

Demoulin (1998) digitised 24 rivers of third to fifth order from 1:25,000 maps, using contour crossings every 2.5 or 5 m. The digitised lengths matched field-levelled lengths to within 0.1% on average. Profiles were resampled every 200 m, smoothed and normalised, so that rivers of different length and relief could be compared (Fig. 2a). Three parameters describe profile shape: E_r, the area between the profile and a straight line joining source and mouth, as a percentage of the triangle below that line; H_{max}, the largest vertical gap between profile and line; and E_q, the normalised distance from the source at which this gap occurs. A well-graded, strongly concave profile has a high E_r and H_{max} and a low E_q.

The key methodological step was to estimate how much of the river gradient is simply explained by hydraulics. Bankfull discharge in the region scales closely with basin area ($r = 0.94$ for 40 rivers), and the mean height of the valley sides, H_{vm}, was proposed as a map-based proxy for bedload supply. H_{vm} correlated with the median bedload size at the river mouth ($r = 0.79$, 18 rivers) but not along individual rivers. The product $H_{vm} \cdot L/A$, which expresses the total valley-side surface relative to basin size, explained most of the variation in mean gradient (Fig. 5). Rivers that plot far from this regression were taken to be influenced by non-hydrodynamic factors, mainly lithology or tectonics.

4.2. Gradient, mountain-front and valley indices (Upper Rhine Graben)

Peters and Van Balen (2007) first compiled a series of six palaeogeographic maps from the Late Miocene to the present, based on earlier work and new terrace and fault mapping.

Table 1. Profile of the three reviewed studies.

	Demoulin (1998)	**Peters and Van Balen (2007)**	**Demoulin (2011)**
Journal	*Geomorphology* 24	*Global and Planetary Change* 58	*Geomorphology* 126
Area	Ardenne massif and forelands (Meuse basin)	Northern Upper Rhine Graben and shoulders	Western Rhenish shield (Ardennes to Palatinate)
Tectonic style	Broad uplift, NE Ardenne strongest; minor faulting	Rift with border and intra-graben faults	En-bloc uplift migrating from S to N
Typical rates	$\leq$0.025 to $\sim$0.1 mm a^{-1}	0.01–0.08 (mean 0.03) mm a^{-1}	up to $\sim$0.5 mm a^{-1} (0.73–0.4 Ma)
Data	1:25,000 maps, 2.5–5 m contours; 24 rivers, orders 3–5, 21–65 km	DGM50 (30 m) and DGM5 DEMs; fault and terrace maps; 20 m contour interval	SRTM 3″ DEM; 85 rivers, orders 2–6, $\sim$15–4000 km^2
Main indices	Gradient, E_r, H_{max}, E_q, H_{vm}, $H_{vm}{\cdot}L/A$	SL, S, S_{mf}, V_f, fault–drainage orientation	$\int H_b$, $\int H_n$, $\int H_r$, I_b, I_r, R_1, S_r
Statistics	Correlation, regression, K-means clustering	Regional means and ranges	Regression on ln$A$, ANCOVA tests
Main question	Where are areas of differential vertical movement?	Which fault segments are active, and since when?	When did uplift affect each area, in relative terms?

They then compared the orientations of 156 intra-graben and 94 shoulder fault segments with 412 stream segments to detect structural control of drainage. Finally, they computed four indices from a 30 m DEM, using a finer DEM along the Western Border Fault (WBF) scarp: the stream length–gradient index SL of Hack (1973), with a 20 m contour interval; its first derivative S, which highlights where SL changes most sharply; the mountain-front sinuosity S_{mf}, over 15 km segments of the border faults; and the valley-floor width-to-height ratio V_f, measured 1 km upstream of the border faults (Bull and McFadden, 1977).

4.3. *Hypsometric integrals at three levels (western Rhenish shield)*

Demoulin (2011) reasoned that a relative fall in base level reaches the landscape in three stages: the trunk stream first, then the drainage network, and finally the hillslopes of the whole basin. Each stage is described by an integral of a normalised elevation curve (Fig. 2b): the classic basin hypsometric integral $\int H_b$ (Strahler, 1952); $\int H_n$, the same integral for the pixels of the drainage network; and $\int H_r$, the integral of the trunk-stream long profile scaled to the basin relief. Differences between them measure how deeply the network is cut into the basin ($I_b = \int H_b - \int H_n$) and how far incision has spread from the trunk into the network ($I_r = \int H_n - \int H_r$). Their ratio, $R_1 = I_r/I_b$, should cancel much of the lithological effect, which acts on both terms. The indices were computed for 85 rivers from the SRTM 3″ DEM. Tests suggested that the coarse resolution changes R_1 by no more than a few per cent.

4.4. *Different data, different questions*

The three studies differ as much in their data as in their aims. The map-based analysis of 1998 used contour intervals of 2.5–5 m, finer than either DEM-based study, but could only sample 24 rivers. The URG study used a 30 m DEM with a 20 m contour interval, which its authors note may be too coarse to detect small gradient changes on the low-gradient graben floor. The 2011 study accepted the lowest resolution of the three in exchange for a nearly universal data source and a large sample of basins. These choices follow from the questions asked: *where* differential movement occurs (1998), *which* fault segments are active (2007), and *when* uplift happened in relative terms (2011).

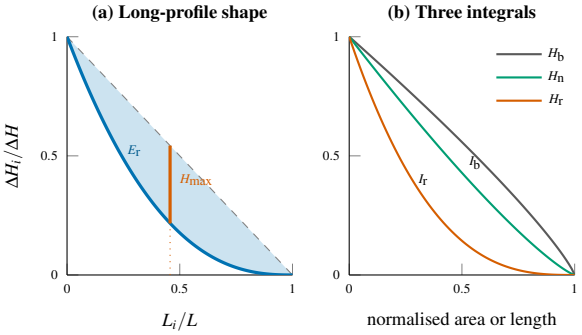


Figure 2. Geometry of the profile and basin indices. (a) Idealised normalised long profile with the parameters of Demoulin (1998). (b) Idealised curves with the same integrals as the Ahr basin example of Demoulin (2011) ($\int H_b = 0.543$, $\int H_n = 0.450$, $\int H_r = 0.263$, giving $I_b = 0.093$, $I_r = 0.187$ and $R_1 = 2.01$).

5. Main findings

5.1. *Ardenne: regional uplift and local subsidence from profile parameters*

The regional means already show a clear pattern (Table 3). Mean river gradient in the uplifted NE Ardenne is twice that of the western massif and nearly three times that of the two forelands (derived ratios 2.0 and 2.8–3.0). The southern foreland, a stable area with weak Mesozoic rocks, has the most concave profiles (E_r 45.9%).

Profile shape gives a finer picture. Clustering on H_{max} and E_q alone produced four recognisable profile types (Fig. 4). Cluster 1 holds well-graded, concave profiles, mostly outside the massif. Cluster 2 groups the slightly less concave rivers of west-central Ardenne. Cluster 3 holds rivers with a convex lower course, and cluster 4 rivers with nearly straight profiles. For the Hoyoux, Samson and Bocq, which flow to the Meuse between Dinant and Huy, neither lithology nor hydrology explains the convexity, and the Meuse itself has not incised there for about 100 ka. Demoulin (1998) therefore inferred subsidence of the Namur area relative to the Condroz, which agrees with repeated levelling surveys from 1948–1978. Straight profiles of the Helle and Salm, by contrast, reflect resistant bedrock combined with uplift of their NE Ardenne source areas.

The regression of gradient on $H_{vm} \cdot L/A$ is the most informative single result (Fig. 5). Except for the Ton, a sand-bed

Table 2. Morphometric indices used in the reviewed studies.

Index	Definition	Reading	Source
Mean gradient	Elevation drop over river length (‰)	Higher where profiles are steep or not yet graded	D98
E_r	Area between profile and source–mouth line, % of triangle	High: concave, graded; low or negative: straight or convex	D98
H_{max}, E_q	Largest profile–line gap; its normalised distance from source	High H_{max}, low E_q: concavity migrated far upstream	D98
H_{vm}, $H_{vm} \cdot L/A$	Mean valley-side height; times river length over basin area	Proxy for bedload/discharge ratio; residuals flag non-hydraulic controls	D98
SL	$(\Delta H/\Delta L) \cdot L$, where L is measured upstream to the source	High: local steepening of the channel	Hack (1973); P&vB07
S	$(\Delta\text{slope}/\Delta L) \cdot L$	Sign changes locate breaks in slope	P&vB07
S_{mf}	L_{mf}/L_s: front length along the foot over straight length	≈1: straight, fault-controlled front; >2: embayed, eroded front	Bull and McFadden (1977); P&vB07
V_f	$2V_{fw}/[(E_{ld} - E_{sc}) + (E_{rd} - E_{sc})]$	Low: V-shaped valley, active incision; high: broad floor	Bull and McFadden (1977); P&vB07
$\int H_b$, $\int H_n$, $\int H_r$	Integrals of normalised basin, network and trunk-profile curves	Long-, middle- and short-term response to base-level fall	D11
I_b, I_r	$\int H_b - \int H_n$; $\int H_n - \int H_r$	Incision of network into basin; spread of incision into network	D11
R_1, S_r	$R_1 = I_r/I_b$; S_r = slope of $R_1 = S_r \ln A + b$	S_r decreases with time since uplift	D11

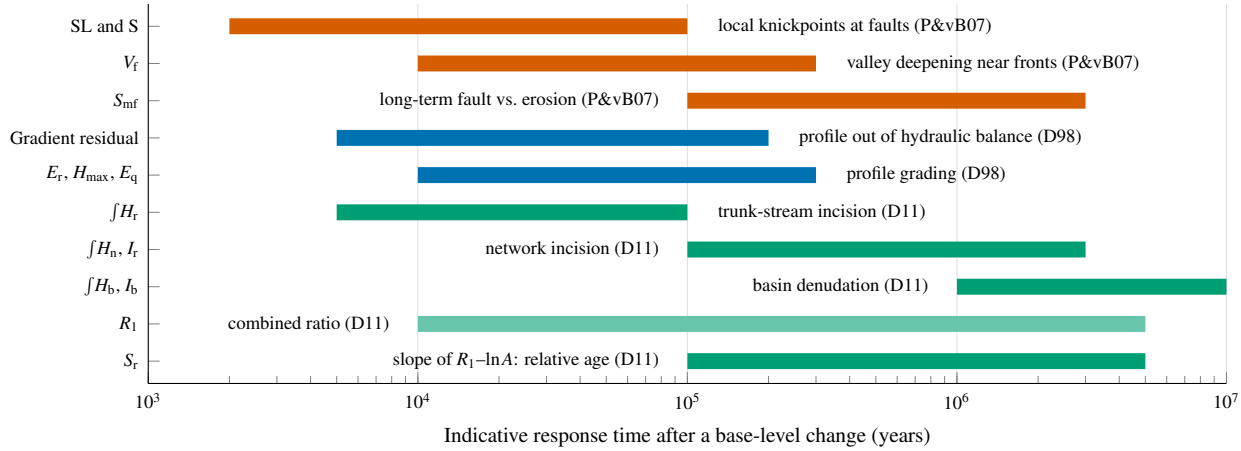


Figure 3. Indicative time windows during which each index reacts to a fall in base level, compiled from the reasoning of the reviewed papers (colours: vermilion, Peters and Van Balen, 2007; blue, Demoulin, 1998; green, Demoulin, 2011). Ranges are qualitative and depend on basin size, rock strength and uplift rate.

Table 3. Mean profile parameters by sub-region of the Ardenne (from Demoulin, 1998).

Sub-region	n	Grad. (‰)	E_r (%)	H_{max}	E_q	$H_{vm} \cdot L/A$
NE Ardenne	6	12.4	17.8	0.17	0.36	1.51
Central Ardenne	5	6.9	23.7	0.19	0.42	0.97
W Ardenne–Condroz	7	6.1	13.2	0.13	0.39	0.90
N foreland	3	4.4	25.0	0.23	0.27	0.51
S foreland	3	4.1	45.9	0.36	0.23	0.66

river on erodible rock, all outliers lie in the NE Ardenne. The Hoegne, Helle and Aine are steeper than their hydraulic setting predicts, which points to delayed grading under current uplift. The Lienne and Salm are less steep than predicted, which suggests strong incision in response to an earlier phase of uplift. Knickpoints along the Lesse, Lomme and Wamme point to a Late Pleistocene base-level fall in the Famenne, which could be either periglacial or tectonic.

Two cautions run through the study. First, the clusters do not follow the tectonic sub-units neatly: rivers of the NE and W Ardenne fall into all four groups, and the very different profiles of neighbouring NE Ardenne rivers remained unexplained. Second, some contrasts are inherited rather than active. The difference between Ardenne and Condroz rivers reflects an Oligocene planation surface cut more than 100 m below the Ardenne surfaces, not present-day movement.

5.2. Northern Upper Rhine Graben: segmented border faults and a preserved tectonic landscape

The palaeogeographic synthesis shows that uplift of the western graben shoulder shaped the Rhine and its terraces throughout the Pleistocene. Offset terraces give deformation rates of 0.01–0.08 mm a^{-1} for the Early–Middle Pleistocene and about 0.03 mm a^{-1} for the Late Pleistocene, so the rate has stayed low and fairly constant (Peters and Van Balen, 2007). Faults with a morphological expression mostly strike NNE, parallel to the graben. Several of them coincide with terrace scarps, which the authors interpret as composite scarps of fluvial and tectonic origin. NW–SE faults appear to guide some

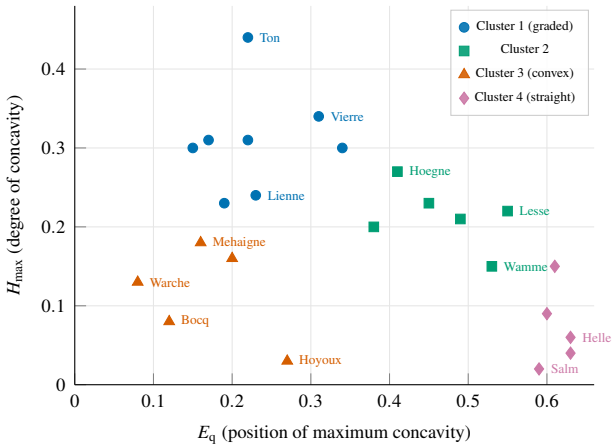


Figure 4. Concavity (H_{max}) against position of maximum concavity (E_q) for the 24 Ardenne rivers, grouped by the K-means clusters of Demoulin (1998). Redrawn from the values in his Tables 1 and 4; only some points are labelled.

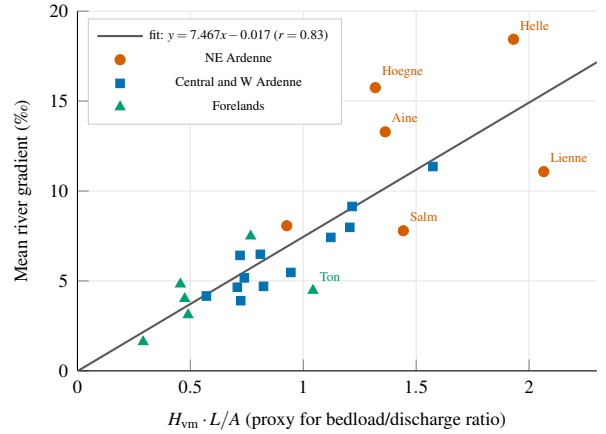


Figure 5. Mean gradient against the bedload/discharge proxy for the 24 rivers of Demoulin (1998), redrawn from his Table 1. Points above the line (Hoegne, Helle, Aine) are steeper than hydraulics alone explains; points below (Lienne, Salm, Ton) are more deeply incised.

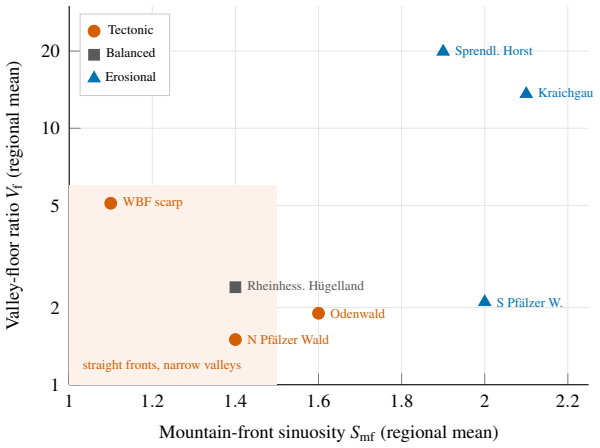


Figure 6. Regional means of mountain-front sinuosity and valley-floor ratio along the northern URG border faults, redrawn from Table 1 of Peters and Van Balen (2007). Classes follow their interpretation.

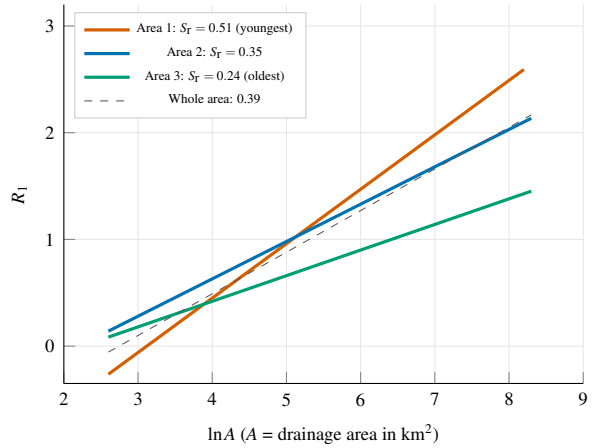


Figure 7. Regional regressions $R_1 = S_r \ln A + b$ for the three uplift-age areas of the western Rhenish shield, drawn from the equations published by Demoulin (2011). Lines are plotted over the basin-size range of the data set. The slope decreases with the age of uplift.

streams in the eastern graben. Linear ENE–WSW scarps in the Vorderpfalz do not follow any known fault and were judged to be purely erosional.

The indices separate the border faults into segments of higher and lower activity (Table 4; Fig. 6). The WBF scarp, the northern Pfälzer Wald and the southern Odenwald show straight fronts, narrow V-shaped or hanging valleys and sharp gradient changes where streams cross the faults. The Kraichgau, the Sprendlinger Horst and the southern Pfälzer Wald show embayed fronts, broad valleys and graded streams. Two comparisons make this result convincing. In the Pfälzer Wald, the north and south are cut in the same Buntsandstein sandstone, yet their indices differ strongly, so rock strength cannot explain the contrast. In the Odenwald, a straight front and V-shaped valleys could be due to resistant granite, but the streams in the sandstone area south of Heidelberg still show strong gradient changes. That segment also lies next to the deepest Quaternary depocentre (about 380 m) and the highest shoulder topography (566 m).

Along the 20 km WBF scarp, longitudinal profiles change from almost concave in the south to convex in the north, and the dry, hanging valleys are not adjusted to the 50–100 m height of the scarp. Because fluvial erosion decreased sharply after about 15 ka, while slow faulting continued, the authors argue that this tectonic landscape is largely a preserved Pleistocene feature rather than proof of present activity. Intra-graben faults, though visible in scarps and in drainage orientation, produced no measurable gradient change at a 20 m contour interval. This is a resolution limit rather than evidence of inactivity.

5.3. Western Rhenish shield: relative uplift age from R_1 and S_r

No single integral or difference index distinguished all three uplift-age areas (Demoulin, 2011). Variation within each area was large, and most of it came from basin size. Once R_1 was regressed on ln$A$, however, the three areas separated clearly (Table 5; Fig. 7). Correlations were high ($\rho = 0.88$–0.93), and covariance analysis showed that the regional regressions differ in both position and slope. The slope S_r falls from 0.51 in the currently rising north, to 0.35 in the area uplifted around 0.8 Ma, and to 0.24 in the area where uplift culminated before

Table 4. Regional means of the geomorphic indices of the northern URG (from Peters and Van Balen, 2007, their Table 1), with the interpretation given in that study.

Region	**|S|**	**SL**	**SL near fault**	V_f **mean**	V_f **range**	S_{mf} **mean**	S_{mf} **range**	**Interpretation**
Eastern graben shoulder								
Kraichgau	1.4	36	34–79	13.6	4.1–31.3	2.1	1.8–2.4	Erosion dominant; graded streams
Odenwald	12.0	94	170–271	1.9	0.3–5.6	1.6	1.2–3.0	Tectonic in south; front shaped by strong rock
Sprendlinger Horst	0.4	31	16–33	19.9	14.7–27	1.9	1.5–2.5	Erosion dominant
Western graben shoulder								
WBF scarp	8.1	76	170–302	5.1	3.4–11.0	1.1	1.1–1.2	Tectonic and fluvial (composite scarp)
Rheinhess. Hügelland	7.8	74	71–101	2.4	1.1–4.6	1.4	1.1–1.8	Erosion and uplift in balance
N Pfälzer Wald	13.4	102	180–442	1.5	0.6–4.0	1.4	1.2–1.9	Active uplift
S Pfälzer Wald	1.3	52	49–100	2.1	0.7–3.5	2.0	1.7–2.3	Erosion dominant
Northwestern regions								
Rhenish Massif	13.0	136	–	–	–	–	–	Gradients reflect resistant rock
Mainz Basin	2.6	63	–	–	–	–	–	Soft rocks, gentle slopes

Table 5. Regional statistics of R_1 in the western Rhenish shield (from Demoulin, 2011, his Tables 2 and 3 and Fig. 3). Area 1: N Ardennes–N Eifel; area 2: central Ardennes–S Eifel–N Hunsrück; area 3: S Ardennes–S Hunsrück–Saar-Nahe. Timing refers to the main uplift pulse.

Area (uplift)	n	R_1 (mean±s)	S_r	b	ρ
1 (rising now)	28	0.99±0.73	0.51	−1.59	0.93
2 (∼0.8 Ma)	35	1.03±0.45	0.35	−0.77	0.88
3 (before 1 Ma)	22	0.58±0.34	0.24	−0.54	0.89
Whole area	85	0.90±0.57	0.39	−1.07	0.87

1 Ma. The youngest area thus has about twice the slope of the oldest (derived ratio 2.1).

The interpretation rests on a simple model of how incision spreads. After a base-level fall, the trunk stream cuts down first and fastest, so $\int H_r$ drops, I_r grows and R_1 rises within about 10^4–10^5 years. Over the following million years or so, incision spreads into the tributaries, $\int H_n$ falls, I_b grows and R_1 declines; much later, slow hillslope denudation lowers $\int H_b$ and R_1 rises slightly again. Because knickpoints travel faster in larger basins, this sequence is delayed in smaller ones, and the slope S_r therefore first steepens briefly and then flattens with time. Single-river analyses along the Ourthe, Sûre and Nahe gave similar slopes but were disturbed where a tributary drains a different uplift area, as at the Alzette and Glan confluences. Regional sets of rivers were therefore preferred.

6. Discussion

6.1. A shared logic: remove the dominant control, read the residual

Despite their different indices, the three studies follow the same strategy. Each first identifies the factor that controls most of the variation in its indices, removes or controls for it, and only then looks for tectonics in what is left. In the Ardenne, this factor is hydraulics: the regression of gradient on $H_{vm} \cdot L/A$ explains about 70% of the variance ($r^2 \approx 0.70$, derived), and tectonics is read from the outliers. In the URG, it is lithology: regions are compared within the same rock type, as in the northern and southern Pfälzer Wald, and lithological explanations are ruled out before tectonic ones are accepted. In the Rhenish shield, it is basin size: R_1 is regressed on $\ln A$, and the regional differences in slope are interpreted as time. In each case the tectonic signal is a second-order effect, visible only after the first-order control is accounted for. This is perhaps the most transferable lesson for other low-strain regions.

6.2. Lithology and inherited relief

All three studies treat lithology as the main source of false signals, but they deal with it in different ways. Demoulin (1998) discussed each river in its geological context and showed that some steepened reaches follow rock boundaries, for example where the Hoyoux and Samson pass from limestone onto Lower Devonian sandstone. Peters and Van Balen (2007) compared regions of similar rock and accepted lithological explanations where these were sufficient, as for the Rhenish Massif streams. Demoulin (2011) formulated R_1 as a ratio so that lithological effects cancel, and chose an area of fairly uniform slaty bedrock. A related problem is inheritance. Both the Condroz planation surface in the Ardenne and the preserved Pleistocene scarps of the URG show that a landscape can look tectonic long after the forcing has slowed or stopped. Morphometry alone therefore cannot prove present-day activity; it needs independent evidence such as seismicity, geodesy or dated deformed deposits.

6.3. From where to when

The three studies also show how the field has moved from mapping patterns to estimating timing. Profile and front indices respond quickly, within about 10^4–10^5 years (Fig. 3), and are well suited to locating active structures or uplifting blocks. They say little about when the movement started. The R_1–S_r approach adds a time dimension by using the fact that different parts of a catchment respond at different speeds. The relation is still relative and qualitative (S_r decreases with time since uplift), and Demoulin (2011) noted that uplift rate and uplift age cannot yet be fully separated. A later study by the same author extended the method towards morphometric dating (Demoulin, 2012). Calibration against areas of known uplift age remains the main step needed to make it quantitative.

6.4. Consistency across the region

Where the study areas overlap, the results agree. The NE Ardenne stands out in 1998 for its steep gradients and for the rivers that plot above the hydraulic regression, and the same area belongs to the youngest-uplift group with the highest S_r in 2011. Cosmogenic dating of the Younger Main Terrace on the Meuse, Ourthe and Amblève supports this picture: post-terrace incision began about 0.73 Ma on the Meuse but only about 0.39 and 0.22 Ma on its Ardennian tributaries, consistent with an erosion wave moving into the massif (Rixhon et al., 2011). Channel steepness and convexity in third-order Ardenne catchments likewise show that these basins are still adjusting (Sougnez and Vanacker, 2011). On the eastern side, the high gradient indices of the Rhenish Massif streams in the URG study agree with the stronger Quaternary uplift of the massif relative to the Mainz Basin, although Peters and Van Balen (2007) attributed the local anomalies mainly to resistant rock.

6.5. Data resolution and sample size

Resolution affects each method differently. Map-based profiles with 2.5–5 m contours capture small gradient changes but are laborious, which limits samples to tens of rivers. A 30 m DEM with 20 m contours misses subtle steps on the graben floor, as Peters and Van Balen (2007) acknowledged. Integral indices are less sensitive to resolution because errors partly cancel, which allowed Demoulin (2011) to use SRTM data. Modern high-resolution DEMs and lidar, together with automated tools for stream-profile analysis (Schwanghart and Scherler, 2014), would now allow the detailed approach of 1998 to be applied to hundreds of rivers. They would also allow the integral approach to be tested against channel-steepness and χ-based methods (Wobus et al., 2006; Perron and Royden, 2013).

6.6. Strengths and limitations

Table 6 summarises the strengths and weaknesses. None of the studies uses absolute ages directly in its morphometric analysis, and all rely on terrace chronologies or geodetic data for interpretation. The 1998 and 2007 results are mostly spatial, while the 2011 result is temporal but only relative. Statistical treatment ranges from correlation and clustering to formal tests of regression equality, and this progression is itself a useful development.

6.7. Towards an integrated workflow

Fig. 8 proposes a workflow that combines the strengths of the three approaches. The geological setting is described first, so that lithological and inherited features are known in advance. Profile indices are then computed for many rivers from a high-resolution DEM, and the dominant non-tectonic control is modelled and removed. Residuals are mapped to show where tectonic influence is likely. Where faults bound the relief, front and valley indices are added. Basin-scale ratios such as R_1 are then used to estimate relative timing. Finally, the results are tested against independent ages and fault data. Composite indices that combine several measures, such as the index of relative active tectonics of El Hamdouni et al. (2008), could serve as a summary at the regional scale.

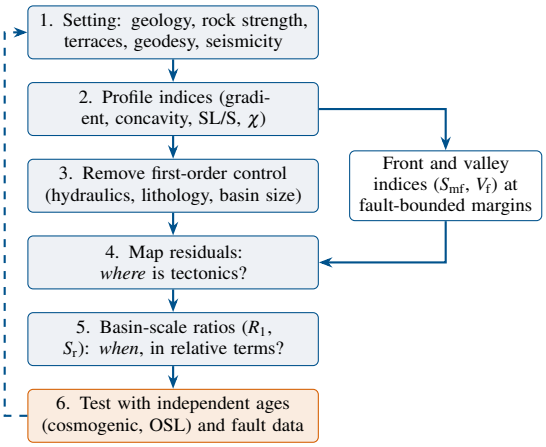


Figure 8. Proposed multi-scale workflow combining the strengths of the three reviewed approaches.

7. Conclusions

The three studies reviewed here show that tectonic signals can be recovered from fluvial landscapes even where deformation is very slow, provided that the dominant non-tectonic controls are dealt with first. In the Ardenne, profile parameters and a map-based bedload proxy identify current uplift of the NE massif, relative subsidence around Namur and a Late Pleistocene base-level fall in the Famenne. In the northern Upper Rhine Graben, the combined use of gradient, front and valley indices separates active segments of the border faults (WBF scarp, northern Pfälzer Wald, southern Odenwald) from segments dominated by erosion. In the western Rhenish shield, the slope S_r of the relation between R_1 and basin size orders three areas by uplift age, from 0.51 in the youngest to 0.24 in the oldest.

Profile and front indices tell us *where* tectonics acts; basin-scale ratios begin to tell us *when*. All of these methods remain sensitive to lithology, inheritance and data resolution, and none yet gives absolute ages. Combining them within a single, multi-scale workflow that is tested against independent chronologies offers the best route to a reliable reading of tectonics from rivers in low-strain intraplate regions.

8. Future research directions

Building on the conclusions above, six lines of work would most improve the reading of tectonics from fluvial landscapes in low-strain regions:

1. **Calibrating S_r.** Apply the R_1–S_r method in regions of known uplift age and rate, ideally with cosmogenic or terrace chronologies, to turn the relative scale into an absolute one.
2. **Separating uplift rate from age.** Test regions where rate varies but timing is similar, and vice versa, as suggested by Demoulin (2011).
3. **High-resolution re-analysis.** Re-examine the Ardenne profiles and the URG intra-graben faults with lidar-based DEMs and contour intervals well below 20 m.
4. **Linking with χ and steepness.** Compare profile, integral and χ-based indices on the same rivers to test which best separates tectonic and lithological signals.

Table 6. Strengths, limitations and open questions of the three approaches.

	Demoulin (1998)	**Peters and Van Balen (2007)**	**Demoulin (2011)**
Strengths	Fine vertical detail from maps; field-validated bedload proxy; explicit separation of hydraulic and other controls	Combines long-term geological record with present landscape; natural lithological controls; segment-scale result	Large sample; lithology reduced by ratio form; adds relative timing; simple, reproducible workflow
Limitations	24 rivers only; proxy not valid along a river; clusters do not match tectonic units; some anomalies unexplained	20 m contours miss intra-graben signals; indices largely qualitative; tectonic landscape may be inherited	Coarse DEM; uplift age and rate not separated; relation $S_r = f(t)$ uncalibrated; basin-shape outliers removed
Open question	Why do neighbouring NE Ardenne rivers differ so much?	Are intra-graben faults still active at the surface?	Can S_r be converted into absolute time?

5. **Quantifying rock strength.** Replace qualitative lithological classes with measured rock strength or erodibility, so that lithology can be removed statistically.
6. **Joint use with geodesy and seismology.** Compare morphometric anomalies with modern GNSS velocities and seismicity to separate active from inherited landscapes.

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Competing interests: The authors declare no competing interests.
Data availability: All values discussed are taken from the published sources cited; the figures were redrawn from these values.

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